



Software Operation & Communication Specification

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Project **Dallas and Charlotte Hub**

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Version

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0.2	9/15/2023	Andrew Rears	Added details for SC functionality.
0.3	9/29/2023	Andrew Rears	Split the SC functionality from the Functional Specification into a separate document (Software Operation & Communication Spec.)
1.0	10/30/2023	Andrew Rears	First version, updated based on feedback.



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1 Introduction

This functional specification provides an overview of the functionality of the BEUMER Group sorter for the DHL Dallas and Charlotte facilities as part of the DHL eCom Mid-Tier projects.

BEUMER has analyzed the concept and attempted to provide a comprehensive description of the proposed functionality, although some details may be incomplete and will require further clarification before the final system design is complete. The current description is by no means definitive; we are always looking forward to an open and constructive dialogue on the final implementation, to ensure that the final solution meets the customer's unique operational needs.

To the extent these sections provide specific quantities or other quantitative measures, it is exclusively provided as support for the narrative. The same applies to illustrations that in most cases should be seen as conceptual. In case of any discrepancy, the proposal scope document sections (Scope of Supply) have priority.

1.1 DHL specifications

The BEUMER Group sorter is a standard machine that for this project is configured for DHL's specification in the area of controls and discharge configuration.

The following documents provide additional adjacent information about the sorter system:

- 623-010008_DHL_eComm_Camera_Specification
- Sorter_Controls_CMC_DHL_eComm
- System_Description_DHL_eComm DFW & CLT
- PS_Conveyor Controls_DHL_eComm DFW & CLT
- 623-010008_BGFusion_Manual
- 623-010008_IP_DHL_eComm
- 623-010008_Account_DHL_eComm
- 623-010008_HOST_Interface_DHL_eComm

1.2 Nomenclature

Nomenclature		
SC	Sorter Control	The PC based upper level logical sorter control system communicating to the HOST
MC	Machine Control	The PLC based machine control system for the

		sorter
Chute		A physical discharge point on the sorter
Destination		The final unique sort destination
Discharge Point		One of possibly two discharges to a destination
ES		Emergency Stop
HOST		Equipment deciding sort destinations
PE		Photo Electric cell (Photo Eye)
Cross-belt		Name for the cross-belt used on the sorter

Table 1-1: Nomenclature

2 Description of Operation

2.1 Sortation

The destination assignment of items is done by the HOST via the [communication interface](#) to the BEUMER sorter control system.

The HOST may assign up to three alternative discharge points on the sorter. If there are duplicate destination entries, the lower-priority (tertiary < secondary < primary) duplicate entries will be ignored.

There are two sort modes:

- a) Sort Plan controlled
- b) HOST controlled

Sort mode can be defined in BG Fusion for each induction.

In a) the sorter control system sends the barcode data to the HOST and receives an "AutomationSlot", in effect a sort code. This is used to lookup a destination in the ASF sort plan.

In b) the sort code is derived from the barcode. Here, other sort plans are used, like TicketToRide, etc.

All sort plans are downloaded from the Master Data Server. Multiple plans may be downloaded and stored and one of each can be activated via BG Fusion.

Summary

'MC' = Beumer Machine Control

'SC' = Beumer Sorter Control

HOST = DHL NewOps

- Individual items are scanned on the sorter and/or on the induction
- The scan result (and weight & volume, where applicable) is transferred to the HOST for each camera system, sorter tunnel or bottom camera system for code conversion.
- The HOST provides a sorter destination and possibly an alternative destination or alternative destinations
- The MC discharges the item into the designated chute
- SC transmits a discharge message to the HOST
- At tray check position (SPS), the MC checks if cross belt tray is actually empty
- SC sends discharge confirmation to the HOST
- The MC clear the tray data so cross belt tray can be available again

2.1.1 Sort Preparation

When using Sort Plan files, prior to the start of sortation, the DHL Master Data server(s) will download any necessary sort plan and chute attribute configuration files to the sorter control system (often referred to as the SC). The sort files are necessary for sorting parcels to pre-determined destinations based on the parcel's extracted sort code. The chute attribute configuration files are necessary for sorting Ticket-to-Ride items only to allowed destinations, based on the parcel's extracted attribute.

Sort plan and chute attribute configuration files may only be created via download to the SC server by the DHL Master Data server(s). The download consists of the Master Data server(s) transferring files onto a shared directory on the SC server (see [section "DHL HOST Interface"](#) for details). The SC server will monitor a particular directory for any newly downloaded files. Upon successful validation, the file will be moved to a "pass" subdirectory. Upon encountering an error in the validation, the file will be moved to a "fail" subdirectory along with an error file containing an error description. Both subdirectories are accessible by the Master Data server(s).

Alternatively, when using HOST Controlled sorting, destinations can be assigned to items with prescribed barcodes, set by sending a combination of DRMs (Destination Request Messages) and DAMs (Destination Assignment Messages). More details about both sort plans can be found in the following subsections.

2.1.1.1 Sort Plan Files

The downloaded sort plan files are used during sortation to translate sort codes to chute(s). For all sort plan files, up to three (3) chutes may be specified per sort code. The translation from mail item barcode to sort code is described in the "CAMERA SYSTEM AND VOLUME SCANNERS" document, for each mail item type.

There will be four (4) separate sort plan files to be downloaded to the SC:

1. Ticket-to-Ride Sort Plan for sortation of domestic mail items containing Ticket-to-Ride barcodes
2. USPS eVS Sort Plan for sortation of domestic mail items containing USPS eVS barcodes
3. CONNECT Sort Plan for sortation of cross-border mail items containing cross-border barcodes
4. ASF Sort Plan for sortation of items where the SC triggers a DRM message for the host

Additionally, all sort plan files may contain chute display information, specifying text to be displayed at a chute's display module, where available.

The common sort plan layout consists of zero or more of the following messages, in any order:

SORT message

Sort code	Primary sort destination	Secondary sort destination	Tertiary sort destination
-----------	--------------------------	----------------------------	---------------------------

DISPLAY message

Chute	Text to be displayed
-------	----------------------

Note: The SC server copies successfully-downloaded sort plan files in the "pass" directory for informational purposes only. The actual sort plan files are stored in a separate internal directory, inaccessible by the Master Data server(s) via the shared directory. Therefore, deleting a file within the "pass" subdirectory will not delete the file from the sorter system. Sort plan files may only be deleted from the system by authorized operators via [BGFusion](#).

2.1.1.2 Chute Attribute Configuration Files

The downloaded chute attribute configuration files are used during sortation to determine to which chutes Ticket-to-Ride mail items are permitted to discharge. Ticket-to-Ride barcodes are the only barcodes that contain a two-digit hexadecimal attribute field. Within the chute attribute configuration file, a chute may be configured to allow any combination of 256 different attributes (actual attribute values are '00' – 'FF').

The principle domestic sort table layout consists of zero or more of the following message:

CHUTE message

Chute	List of allowable attributes ('00' – 'FF')
-------	--

Note: The SC server copies successfully-downloaded chute attribute configuration files in the "pass" directory for informational purposes only. The actual chute attribute configuration files are stored separate internal directories, inaccessible by the Master Data server(s) via the shared directory. Therefore, deleting a file within the "pass" subdirectory will not delete the file from the sorter system. Sort plan files may only be deleted from the system by authorized operators via BGFusion.

2.1.1.3 Activation

The sort plan and chute attribute configuration files may be activated, deactivated, and deleted via BGFusion. At most one (1) of each type of sort plan file may be active at the same time. Similarly, at most one (1) chute attribute configuration file may be active at the same time. All four file selections (three sort plan & one chute attribute configuration) are activated at the same time, via BGFusion. Upon activation, the system performs basic validation and also ensures that the domestic sort plans (Ticket-to-Ride & USPS eVS) do not share chutes with the CONNECT sort plan. If validation fails, no changes are made to the currently active files and feedback is given to the operator. The validation may fail for one of the following reasons:

- No active sort plans are selected
- A Ticket-to-Ride sort plan is selected but no Chute Attribute configuration is selected
- The domestic and CONNECT sort plans share chutes

In which case, the user is prompted with a warning message. The user then has the choice to confirm or cancel the activation of the sort plans.

2.1.1.4 HOST Controlled Sorting

BGFusion will include inductions flags called NewOpsSendDRM and NewOpsWaitDAM. If this NewOpsSendDRM flag is disabled, the item will use the Sort Plan File functionality, as described above. If the NewOpsSendDRM flag is enabled, BEUMER will determine the type of barcode (TTR\EVS) and send a DRM request to NewOps with that barcode. The destination assignment is optional for sortation; BEUMER will only wait to use the contents of this message for sortation if the induction has the NewOpsWaitDAM flag enabled.

If the response does not include the destination assignment data, the item will recirculate on the sorter without a destination. If the response does include the destination assignment data, the NewOps provided new sort code will take precedence over the original extracted sort code. The data format of this new sort code will be similar, but not the same as TTR (for example, "V361A01EXP"). BEUMER will use this sort code to look up the destination in a new type of sort plan file and determine the chute destination.

If NewOpsWaitDAM flag is disabled, BEUMER will use the sort code extracted from the barcode found on the item and look up the destination in the loaded TTR\EVS sort plan file.

2.1.2 Barcode Specifications

2.1.2.1 Ticket-to-Ride Barcodes

The ticket-to-ride barcode is a DHL-defined barcode type. There are two barcode constructs in use: customer-generated & DHL-generated (shown below). All ticket-to-ride barcodes are QR codes, with starting character 'V', of barcode lengths 26, 32, or 36, of size 1.5 cm x 1.5 cm. The sorter system identifies a barcode as ticket-to-ride if it is of type QR, has first character 'V', and is of length 26, 32, or 36.

- QR-Code of length 26 containing a V in the first position → Ticket-to-Ride
- QR-Code of length 32 containing a V in the first position → Ticket-to-Ride
- QR-Code of length 36 containing a V in the first position → Ticket-to-Ride
- Any other length would be considered a cross-border barcode

Within the Ticket-to-Ride barcode, the concatenation of the 'DHL Sort Code Version Number', 'DHL Outbound (Destination) Sort Code', and 'DHL Inbound (Final Slot) Sort Code' fields will be used as the sort code during sortation.

Important: The sorter system extracts this sort code by parsing out characters 1 through 8.

Within the Ticket-to-Ride barcode, the concatenation of the 'Bin Constraint Parameter 1' and 'Bin Constraint Parameter 2' fields will be used as the item attribute during sortation. The sorter system extracts this item attribute by parsing out characters 9 through 10.

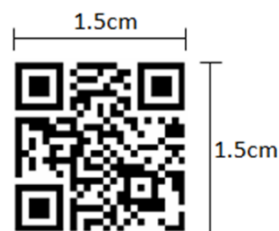


Figure 2-1: QR Barcode Example

Compared to the previous sites provided by BEUMER, the barcode communication scheme now uses the | sign as delimiter, which in affect makes the fields variable length, so the field length in the tables below is now considered the maximum field length value.

Position	Field Description	Field Length
1-2	DHL Sort Code Version Number - Capital "V" + single-digit version number - No spaces	2
3-5	DHL Outbound (Destination) Sort Code Underscore "_" + 2-digit outbound sort code	3
6-8	DHL Inbound (Final Slot) Sort Code Single alpha character + 2-digit number	3
9	Bin Constraint Parameter 1 Single-digit number	1
10	Bin Constraint Parameter 2 Single-digit number	1
11-26	DHL Internal Identifier Number 16-digit internal DHL tracking number	16

Figure 2-2: DHL-Generated Ticket-to-Ride Barcode Elements

The "field length" column now states the maximum field length.

Example: V6|71|A01|0|2|3072607608496710

Character Position	Field Description	Field Length
1-2	DHL Sort Code Version Number - Capital "V" + single-digit version number - No spaces	2
3-5	DHL Outbound (Destination) Sort Code Underscore "_" + 2-digit outbound sort code	3
6-8	DHL Inbound (Final Slot) Sort Code Single alpha character + 2-digit number	3
9	Bin Constraint Parameter 1 Single-digit number	1
10	Bin Constraint Parameter 2 Single-digit number	1
11-32 or 11-36	USPS Confirmation Number (PIC) 22 or 26-digit USPS tracking number	22 or 26

Figure 2-3: Customer-Generated Ticket-to-Ride Barcode Elements

The "field length" column now states the maximum field length.

Example: V6|71|A01|0|2|9274899996327313016349

Note: The sort code is extracted from the barcode.

2.1.2.2 USPS eVS Barcodes

The USPS eVS barcode is a USPS-defined barcode type. There are six barcode constructs in use (shown below). All USPS eVS barcodes are GS1-128, with "420" as the Routing Application Identifier field as shown in the figure below. The sorter system identifies a barcode as USPS eVS if it is of type GS1-128 and starts with "420".

Within the USPS eVS barcode, the 5-digit ZIP code field will be used as the sort code during sortation. The full ZIP code field can be found between the "420"

routing application identifier field and the second FNC1 character, as illustrated below (note: the first FNC1 character is part of the barcode header and will not actually be transmitted by the scanner). The sorter system extracts the 5-digit ZIP code by parsing out the 5 characters immediately following the leading "420" string.

Code	Description	POSTAL ROUTING CODE Field Length		TRACKING NUMBER Field Length					PIC Length	Total Barcode Length
		Postal Routing AI	Dest ZIP	Channel AI	STC	MID	Serial Number	Check Digit		
C01	Commercial Mail - (Nine-digit Mailer ID, 9-digit ZIP Code)	3	9	2	3	9	7	1	22	34
C02	Commercial Mail - (Nine-digit Mailer ID, 5-digit ZIP Code)	3	5	2	3	9	11	1	26	34
C03	Commercial Mail - (Nine-digit Mailer ID, 5-digit ZIP Code)	3	5	2	3	9	7	1	22	30
C05	Commercial Mail - (Six-digit Mailer ID, 9-digit ZIP Code)	3	9	2	3	6	10	1	22	34
C06	Commercial Mail - (Six-digit Mailer ID, 5-digit ZIP Code)	3	5	2	3	6	14	1	26	34
C07	Commercial Mail - (Six-digit Mailer ID, 5-digit ZIP Code)	3	5	2	3	6	10	1	22	30

Figure 2-4 USPS eVS Barcode Constructs on the Sorter

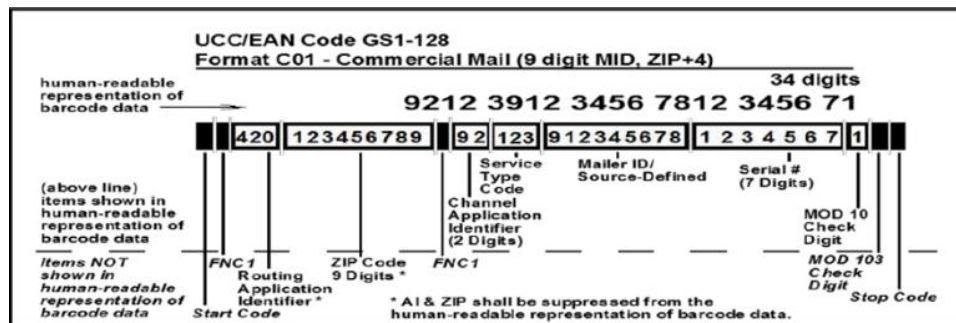


Figure 2-5 Barcode Fields of USPS eVS Barcodes

Note: The sort code is the first 5 digits of the zip code field.

2.1.3 Chute Assignment

If the item is an exception item, it will be sent to the corresponding exception destination (see [section "Sorting Exceptions"](#) for further details regarding sorting exceptions).

Once the sorter control system has scanned a parcel for barcodes, identified the parcel type (e.g. Ticket-to-Ride, USPS eVS, or cross-border), and extracted its sort code (and attribute, if applicable), chutes can be fetched from either the parcel's corresponding sort plan or the Destination Assignment table to be considered for sortation.

For a USPS eVS and cross-border parcel, the corresponding sort plan is searched for an entry whose sort code field that exactly matches the parcel's extracted sort code. An entry may contain up to three (3) chutes to be considered for sortation (primary, secondary, & tertiary).

For a Ticket-to-Ride parcel, the Ticket-to-Ride sort plan is searched for all entries whose sort code fields are left-substring matches of the parcel's extracted sort code. Each entry may contain up to three (3) chutes to be considered for sortation (primary, secondary, & tertiary). Entries whose sort code matches more of the parcel's extracted sort code are prioritized for chute assignment. Only the two (2) best matches are used in sortation, for a maximum of six (6) chutes to be considered for sortation. Furthermore, only chutes whose chute attribute configuration contains the parcel's extracted attribute will be considered for sortation.

If no chutes can be considered for sortation, the item is deemed a no-data exception item and sent to its corresponding exception destination (see [section "Sorting Exceptions"](#) for further details regarding sorting exceptions).

For all chutes being considered for sortation, the chute assignment order (e.g. order of discharge) is determined by the following:

- Matching sort plan entries whose sort code matches more of the extracted sort code (applies only to Ticket-to-Ride parcels)
- Primary chutes will be considered for assignment over secondary chutes, for the same entry
- Secondary chutes will be considered for assignment over tertiary chutes, for the same entry

Once the chute assignment order is determined, only the highest-priority chute will be used for sortation.

2.1.4 Redirection Rules

If a parcel could not be sorted at its initial assignment, defined rules determine the subsequent behavior.

If a parcel fails to discharge to its initial chute assignment before a configurable number of attempts has been reached (*Recirculation Reassignment* parameter, configurable via BGFusion), the parcel will then simultaneously consider all eligible chutes for discharge. From this point on, a discharge attempt will be made at any eligible chute as the parcel passes the chute.

If a parcel recirculates a configurable maximum number of times (*Max # of Recirculations* parameter, configurable via BGFusion), it will be sent to the corresponding exception destination (see [section "Sorting Exceptions"](#) for further details regarding sorting exceptions).

2.1.5 Sort Logic Change Orders from Previous DHL E-Com Sorters

Since the original release of the DHL EWR eCommerce sorter, change orders have been sold and implemented across the EWR and LAX sorters. Due to the sort logic for this DFW eCommerce sorter being based on aforementioned sorters, the functionality that those change orders brought will be implemented into DFW eCommerce. While this functionality has been specified in other places within this document, their inclusion will be mentioned again inside this section.

- HOST Sort Logic Change
 - This change added a new method of assigning item destinations, aside from the use of sort plan files. More information on this change can be found in [section "HOST Controlled Sorting"](#).
- Weight Scan Interface
 - Inductions scanners will send item weight and volume data to the SC system. This information about items is used to influence sorting decisions for items, such as for the STC Reject/Overlabel functionality.
- STC Reject
 - This change adds a new flag that can be applied to the inductions on BGFusion, which determines if that induction should have the 'Overlabel' filter applied to items which it inducts. Based on the STC code present on the EVS barcodes, items of certain weights will be sent to the STC Reject special destination. More information about this change can be found in [section "Sorting Exceptions"](#).

2.2 Sorter Error Handling

2.2.1 Sorting Exceptions

The item may be sent to an exception destination, in most cases determined by a Parcel. Parcels may experience various sorting exceptions, in which case they will be discharged to the corresponding configured exception destination, unless otherwise stated. Exception destinations can be configured via BGFusion.

- No Read
 - The system can be configured to attempt multiple times based on a configurable *max-scan* parameter (configurable via BGFusion). If a parcel is detected and the scan tunnel returns a scan result containing no barcode, after the configurable maximum number of attempts, the parcel is marked as "No Read" and will be discharged to the configured "No Read" exception destination.
- Multiple Read
 - If a parcel has been detected and a valid scan result is obtained from the scan tunnel, then if the parcel contains multiple of the same domestic barcode (e.g. two Ticket-to-Ride, or two USPS eVS) barcodes, then it is marked as a "Multiple Read" and sent to the configured "Multiple Read" exception destination.
- No Data
 - If a parcel is detected and a valid scan result is obtained from the scan tunnel, then the parcel may be marked as "No Data" if any of the following occur during the chute assignment process:
 - The cross-border barcode was unable to be translated to a sort code since it did not match any of the received cross-border parcel information (applicable only to cross-border parcels). See [section "Communication Interfaces"](#) for details on how cross-border parcel information is transmitted to the sorter.
 - The extracted sort code does not match an entry in the corresponding sort plan.
 - The extracted sort code matches at least one entry in the corresponding sort plan, but the parcel is not eligible to discharge to any of the chutes due to mismatching attributes (applicable only to Ticket-to-Ride parcels).

If the parcel is marked as "No Data", it will be discharged to the configured "No Data" exception destination.

- Discharge Fault
 - If a parcel that should have been discharged to a destination is still detected on the Crossbelt, SC will be notified via telegram; the item will then continue to attempt to discharge at its assigned location. If the amount of these discharge fault telegrams is higher than the parameter set in BGFusion, then it is marked as "Discharge Fault" and will be discharged to the configured "Discharge Fault" exception destination. Occupation sensors that detect whether a belt tray is occupied are located at the end of each chute bank, per sorter.
- Stray Item
 - Prior to an induction area a SPS is installed, if the SPS unexpectedly detects a parcel on a Crossbelt (e.g. the parcel was not assigned to that Crossbelt at the induction), then it is marked as a "Stray Item" and will be discharged to the configured "Stray Item" exception destination
- Too Many Recirculating Trays
 - If the total number of recirculating Crossbelts exceeds a configurable percentage (*Recirculating Tray %* parameter, configurable via BGFusion), then recirculating parcels are marked with the "Too Many Recirculating Trays" exception and will be discharged to the configured "Too Many Recirculating Trays" exception destination. Once the total number of recirculating trays drops below the configurable percentage, recirculating trays are no longer subject to this exception.
 - There is also a "Recirculating Tray %" parameter tracked by SC, which is used to stop the inductions from inducing items onto the sorter if the total number of recirculating trays exceeds the parameter value. The inductions will resume inducing items onto the sorter once the total number of recirculating trays drops below the parameter value.
 - **Note:** If the total number of recirculating trays exceeds the configurable percentage, parcels that have not yet recirculated will not be marked with this exception; only upon completing at least one recirculation will the parcel be subject to this exception.
- Tray Spacing
 - Due to necessary delays when identifying chutes as FULL, consecutive parcel discharges to certain chute types may cause undesirable consequences (e.g. sorter jams, overfilling sacks, parcels reaching sorter track, etc.). As such, the sorter control system may need to limit consecutive discharges to particular chute types to a certain number of trays, and then wait to resume

consecutive discharges after skipping a certain number of trays. This feature is known as tray spacing. The actual number of trays that may discharge consecutively and the actual number of trays that are skipped before further consecutive discharges may resume are determined by physical tuning during on-site commissioning.

- If a parcel cannot discharge to a chute due to tray spacing, it will follow the regular redirection rules.
- Overlabel/STC Reject
 - Inductions can be assigned the STC Reject flag inside of BGFusion; this flag gives each item that passes through the induction an additional set of requirements to pass before it can be sorted normally.
 - Based on the STC code extracted from the EVS barcode, it will be sorted based on the below table.

STC Code	Logic
021, 419, 422	Do Not Reject
001, 703, 704, 748, 835	Reject if item weight is greater than or equal to 1lb
055, 108, 612, 615	Reject if item weight is less than 1 lb
All other STC codes	Reject All

- If the item is rejected, it will go to the Overlabel special destination.
- If an item is inducted from an STC flagged induction, and it has an unknown weight OR fits any other exception category (no read, multi-read), it will be redirected to the Overlabel special destination.

2.2.2 Operational Exceptions

The following statements describe the standard methodology of operational exceptions.

- Oversized Items
 - Items exceeding the maximum item length and width of the sorter will be reverted into a reject bin. If bin is full, then the induction will need manual intervention. Please see the Sorter Controls document for product size specifications.
- Overweight Items
 - Units exceeding the maximum item weight of the sorter will be reverted into a reject bin. If bin is full, then the induction will need manual intervention. Please see the Sorter Controls document for product weight specifications.

- Disable / Enable Tray
 - If an occupied tray/belt is manually disabled by means of the MC, the corresponding item will not be able to be discharged for the duration of the disabled carrier, meaning the assignment of the item should be reset by the HOST. Normally, carriers will be disabled after a mechanical defect, so it is not possible to discharge the item.
- The operator must remove the item from the broken carrier and induct it again. If the item is left on the carrier, it will continue to recirculate with the sorter, but MC will not request a destination from Host until the carrier is unblocked. When the carrier becomes enabled, the item will then be reported as a stray item to Host.
 - A discharge confirmation message with a dedicated reason code will be send to the HOST.
- Disabled Destination
 - A disabled destination means that the sorter control system assumes the destination is electrically disconnected. The lights, valves etc. will not be activated from the control. If a destination is disabled by SC, shipments for that destination will report failure to discharge to the HOST. The HOST will be able to re-assign to an exception destination or another chute spontaneously or upon request.

2.3 Chute Displays – ATOP Functionality

The system includes PTL displays to inform and alert the operators. There is a 5-character 16-segment display per destination, one display per chute.



Figure 2-6: AT705E 5-digit PTL Unit

Besides the numeric display, these units have a lighted push-button (with different colors) and two function keys.

The AT500 controllers are connected to the Ethernet network and the sorter control system.

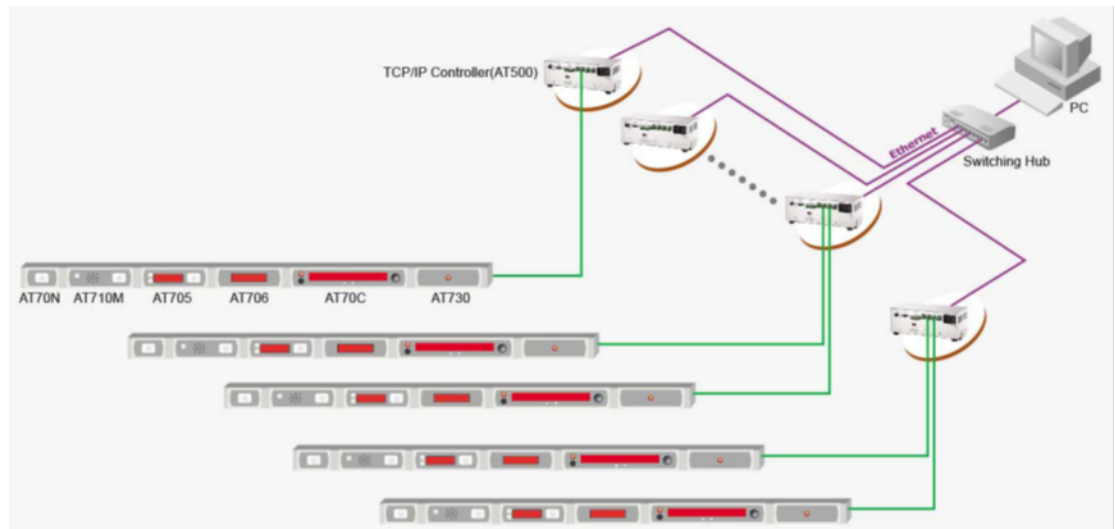


Figure 2-7: Network Architecture for Chute Displays

The displays are mounted in a rail system running along the chute front. They are easily replaced and configurable. All displays are connected to a number of concentrator boxes (AT500) that communicate with the sorter control system via an Ethernet TCP/IP connection.

There are three components on the AT705E display unit:

- A 5-character 16 segment display segment
- One multicolored LED light
- Two smaller pushbuttons

The multi-colored LED light has 6 different light colors: RED, GREEN, YELLOW, BLUE, PURPLE, and LIGHT BLUE.

The 5-character sixteen segment displays shows the currently assigned sort destination as per default (e.g. "EWR" or "LAX"). Once the "up" pushbutton is pressed, the display will show the currently assigned gaylord type. By pressing the "up" pushbutton again, the gaylord type can be changed to the desired type (gaylord type 1 to 3 or no gaylord).



Enabling and Disabling the Destination

The chute can be disabled and enabled from the ATOP display "confirmation" pushbutton.

Selecting Container type

Container type can be selected with ATOP display "up" pushbutton.

Chute Full

When the chute is FULL, the ATOP display will flash yellow, and the destination will be disabled.

Jam Detected

In case of the JAM photo-eye being blocked, the ATOP display will turn on solid yellow light.

3 Communication Interfaces

3.1 DHL HOST Interface

The sorter control system communicates with three separate DHL host systems: [NewOps](#), [CONNECT](#), and [Master Data](#).

The information in this section is an overview of the data and data structures being communicated between the host systems and the sorter control system. Refer to each host system's specific communication protocol document for technical details.

3.1.1 NewOps

Communication between SC and the NewOps system is based on HTTP communication using RESTful APIs. Both SC and NewOps will host a RESTful API as servers to accept client connections from each other. If there is a loss of connection between SC and NewOps, a BGFusion alarm will be raised to alert operators to the broken connection.

SC will communicate the following information to HTTP endpoints via the NewOps RESTful API:

1. Authentication Request (SC → NewOps)
 - This message is sent by the SC system to the host Authentication server to get a valid token.
 - Fields to be transmitted via Authentication Request
 - Client ID
 - Client Secret
 - Grant Type
 - A message should be sent by NewOps as an HTTP response to the Authentication Request from the SC system. This message provides a valid token requested by the SC system.
 - Fields to be transmitted via Authentication Response
 - Access Token
 - Expires In
 - Token Type
 - The access token is then used as a parameter in the "Destination Request" message.
2. Mail Item Sorted (SC → NewOps)
 - This message is sent by the SC system for each mail item that is successfully sorted. An item is deemed successfully sorted by the

SC if the item is confirmed to have sorted to any of its intended destinations listed by the sort plan.

➤ Fields to be transmitted via Mail Item Sorted message:

- Scan Time
- Chute ID
- Barcode (2D/1D)
- Associated Sort Plan
- Induction ID

3. Mail Item Rejected (SC → NewOps)

➤ This message is sent by the SC system for each mail item that is not successfully sorted. An item is deemed not successfully sorted by the SC if the item is confirmed to have sorted to a destination other than any of its intended destinations listed by the sort plan.

➤ Fields to be transmitted via Mail Item Rejected message:

- Scan Time
- Chute ID
- Barcode (2D/1D)
- Reject code
- Reject description

4. Container Closed (SC → NewOps)

➤ This message is sent by the SC system for each chute that is closed so that the associated sack can be packed out by the operator.

➤ Fields to be transmitted via Container Closed message:

- Machine ID
- Chute ID
- Associated sack barcode (if any)

5. Container Status Request (SC → NewOps)

➤ This message is sent by the SC system cyclically every 10 seconds for a list of chutes with the mobile packout flag enabled to query NewOps for the sack barcode to be associated to its respective chute within the SC system. This message is also sent for an individual chute if its green ATOP button is pressed to open the chute so that NewOps can send the associated sack barcode in response.

➤ Fields to be transmitted via Container Status Request message:

- Machine ID
 - List of chute IDs
 - A Container Status message should be sent by NewOps as an HTTP response to the Container Status Request message. This message provides associated sack and threshold information for the chute requested by the SC system.
 - Fields to be transmitted via Container Status message:
 - List of container information, where each container information object contains the following:
 1. Chute ID
 2. Associated sack barcode (empty if none)
 3. Threshold information (empty if none)
6. Destination Request (SC → NewOps)
- This message is sent by the SC system for each mail item scanned on the sorter that has been inducted from an induction with the "Send DRM" induction flag enabled in BGFusion.
 - The Destination Request needs an authentication token in the request parameters. If the authorization information is missing or invalid, the host will not accept the message and respond with HTTP status code 401 (Unauthorized). In that case we will resend the Authentication request to receive a new valid token and then re-send the Destination Request message.
 - Fields to be transmitted via Destination Request message:
 - Item ID
 - Barcode (2D/1D)
 - Width
 - Length
 - Weight
 - Height
 - Dimension Unit of Measurement
 - Weight Unit of Measurement
 - Cuboidal
 - A Destination Assignment message should be sent by NewOps as an HTTP response to the Destination Request from the SC system. This message provides any associated destination assignment information for the item requested by the SC system.

- If the induction on which the item was inducted has the "wait for destination assignment from NewOps" flag disabled, the SC system will use the downloaded sort plan as a default. If NewOps sends a Destination Assignment message, this will overwrite the default sort code extracted from the barcode. SC will then look up the newly-provided sort code in the downloaded sort plan to assign destinations to the item.
- Otherwise, if the flag is enabled on the induction, the SC system will exclusively use the Destination Assignment message to assign the item to a destination. The item will recirculate until it receives a Destination Assignment message from NewOps or until it exceeds maximum recirculations.
- If for some reason, the hosts is unable to process the message due to a business rule, it will respond with a 422 error response. In that case, SC will try to interpret the message body.

NOTE: If the **errorCode** field in the error response message body is not 'null', then SC will treat the errorCode as a sortCode (rejectCode) and will attempt to assign the item to a discharge bin associated with that sortCode in the ASF sort plan.

- Fields to be transmitted via Destination Assignment message:

<u>200 OK</u>	<u>4XX Error Response</u>
AutomationSlot (sortCode)	DetailedUserMessage
Item ID	errorCode
Barcode (2D/1D)	Errors
Width	userMessage
Length	
Weight	
Height	
DimensionUom	
weightUnit	
Cuboidal	

Note: The SC will attempt to preserve sent messages not received by HOST for a certain number of days (configurable via BGFusion, default: 7 days) before purging these messages from the system. While these are preserved, they will be re-sent per defined retransmission rules (described in the NewOps communication protocol document).

3.1.2 CONNECT

Communication between SC and the CONNECT system is based on HTTP communication using RESTful APIs. Specifically, SC will act as server and host a RESTful API for CONNECT to consume as the client.

SC will communicate the following information to HTTP endpoints via the NewOps RESTful API:

1. Consignment (SC → CONNECT)

- This message is sent by the SC system for each mail item scanned on the sorter in accompaniment to the Mail Item message. The consignment number will always be hardcoded to 999999 and the customer name and customer number will always be blank.
- Fields to be transmitted via this message:
 - Consignment number
 - Customer name
 - Customer number

2. Mail Item (SC → CONNECT)

- This message is sent by the SC system for each mail item scanned on the sorter upon receiving the first good-read scan for the item. If the item does not have an assigned destination when it reaches a request point, this message will be sent again by the SC system.
- Fields to be transmitted via this message:
 - Machine ID
 - List of item barcodes
 - Consignment number
 - Width
 - Length
 - Weight
 - Height
 - Cuboidal

CONNECT will communicate the following information to HTTP endpoints via the SC RESTful API:

1. Mail Item Data (CONNECT → SC)

- This message is sent by CONNECT for each encoded mail item that is to be sent to the sorter. The message contents shall contain the mail item's barcode and sort code pairing to be used by the sorter when scanning and sorting the mail item.

- Fields to be transmitted via this message:
 - Barcode
 - Sort code

3.1.3 Master Data

Communication between SC and the Master Data server(s) is based on file transfer over sFTP. Specifically, SC will act as server and host a shared directory for the Master Data server(s) to transfer files for processing.

The Master Data server(s) will transfer the following file types to the SC server:

1. Domestic Ticket-to-Ride Sort Plan File
 - Contains sort code and destination information for sortation of domestic mail items containing Ticket-to-Ride barcodes. May also optionally contain display information to be displayed at chutes with chute displays.
 - Data to be transmitted via this file type:
 - SORT message, containing the following fields:
 - Sort Code
 - Primary Destination
 - Secondary Destination
 - Tertiary Destination
 - DISPLAY message, containing the following fields:
 - Chute
 - Display Text
2. Domestic eVS Sort Plan File
 - Contains sort code and destination information for sortation of domestic mail items containing USPS eVS barcodes. May also optionally contain display information to be displayed at chutes with chute displays.
 - Data to be transmitted via this file type:
 - SORT message, containing the following fields:
 - Sort Code
 - Primary Destination
 - Secondary Destination
 - Tertiary Destination

- DISPLAY message, containing the following fields:

- Chute
- Display Text

3. CONNECT Sort Plan File

- Contains sort code and destination information for sortation of cross-border mail items. May also optionally contain display information to be displayed at chutes with chute displays.

- Data to be transmitted via this file type:

- SORT message, containing the following fields:

- Sort Code
- Primary Chute
- Secondary Chute
- Tertiary Chute

- DISPLAY message, containing the following fields:

- Chute
- Display Text

4. ASF Sort Plan File

- Contains sort code and destination information for sortation of items that SC sent a DRM message to host. May also optionally contain display information to be displayed at chutes with chute displays.

- Data to be transmitted via this file type:

- SORT message, containing the following fields:

- Sort code
- Primary destination
- Secondary destination
- Tertiary destination

- CHUTE INFO message, containing the following fields:

- Chute
- Display text

5. Chute Attributes Configuration File

- Contains attributes per chute to match with Ticket-to-Ride mail item attributes.

- Data to be transmitted via this file type:
 - CHUTE message, containing the following fields:
 - Chute
 - Attribute 1
 - Attribute 2
 - ...
 - Attribute 256

When a new file is transferred and process by SC, it is stored internally. This will not affect any currently active sort plan or chute attribute configuration. These downloaded sort plans and chute attribute configurations may be set to active via the BGFusion by an operator.

4 Sorter Control System

The BEUMER sorter control system is based on a standard framework and architecture common to all project specific implementations making each project supportable by a large number of software and controls engineers. The resources used for a specific project implementation may be global, but the effort will be led by and supported by the BEUMER Corporation's automation group in New Jersey, USA.

4.1 Language

All documentation and software screens are provided in English language.

4.2 BGFusion – Graphical User Interface

The BEUMER sorter control system provides a web server enabling equipment running a Chrome server to access the user interface. This GUI is called BGFusion.

The user interface includes all necessary features to support the operation of the sorter, including:

- Maintaining sort tables
- Viewing equipment status
- Handling and acknowledging alarms
- Viewing production reports
- Setting of system parameters

More specific details about the operation can be found in the BGFusion manual (created upon the completed development of BGFusion for this sorter system).

4.2.1 Equipment Status

The BGFusion includes a local graphical HMI displaying the sorter, inductions, and discharges.

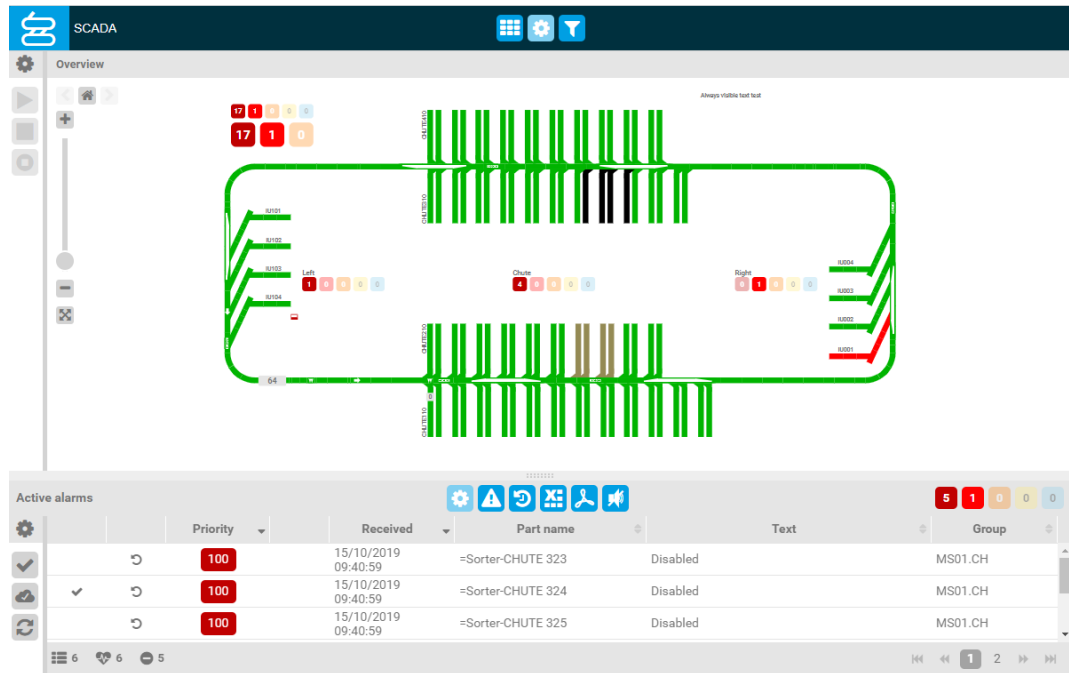


Figure 4-1 - Sample screen of the local BGFusion HMI

Additional screens will display specific sub-systems in greater detail.

4.2.2 Statistics

The BGFusion provides four different main production reports:

- Sorter statistics
- Induction statistics
- Scanner statistics
- Destination statistics

In addition, report on special destinations, and recirculation will be available.

This report displays some general counter of the selected time range.

Selection:

- Date / Time
- Interval

The following counters will be displayed:

Column	Description
Running time	Total running time of the sorter within the selected range
Utilization time	Time with at least one tray occupied with item
Total Items	Sorted Items (discharged) including special cases like No-Read

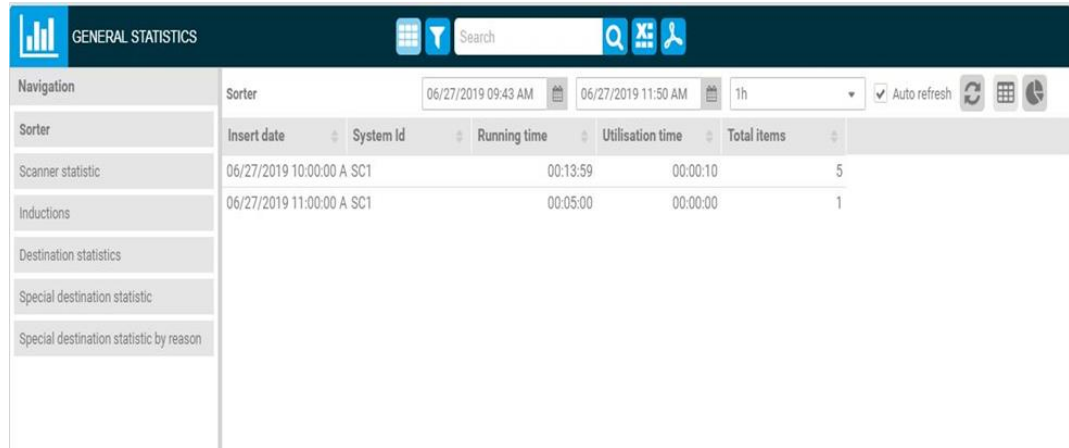


Figure 4-2 - Sorter statistical report

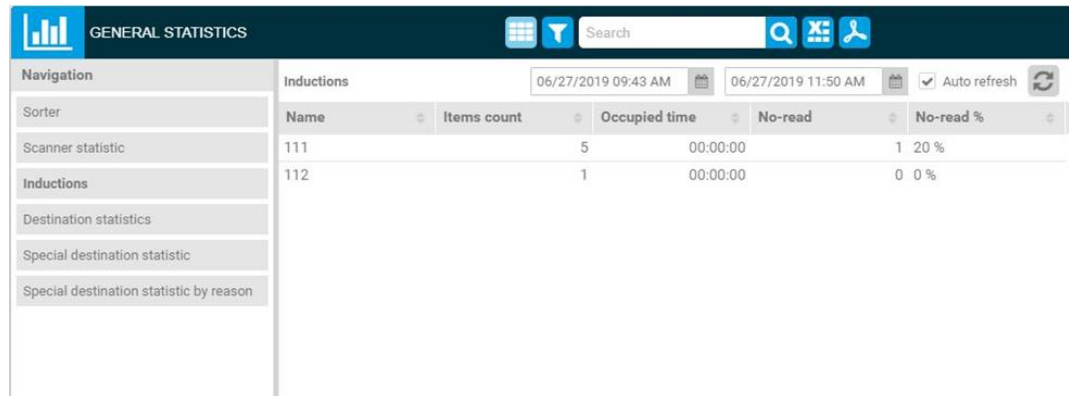
4.2.3 Induction Statistics

The induction statistics indicate the utilization of the induction on a certain time range in a grid.

- Statistics arranged by induction

Selection:

- Date / Time



GENERAL STATISTICS					
Navigation		Inductions			
Sorter		06/27/2019 09:43 AM - 06/27/2019 11:50 AM			
Scanner statistic		Auto refresh			
Inductions		Name	Items count	Occupied time	No-read
Destination statistics					No-read %
Special destination statistic					
Special destination statistic by reason					
		111	5	00:00:00	1 20 %
		112	1	00:00:00	0 0 %

Figure 4-3 - Induction statistical report

The following statistical values are available:

Column	Description
Items	The total amount of inducted items will be displayed
Run-time	Running time of the induct will be displayed
No Read	The total amount of inducted items which were discharged at a special chute due to no reads. The No-read is also shown as a percentage of the total amount of inducted items

4.2.4 Scanner Statistics

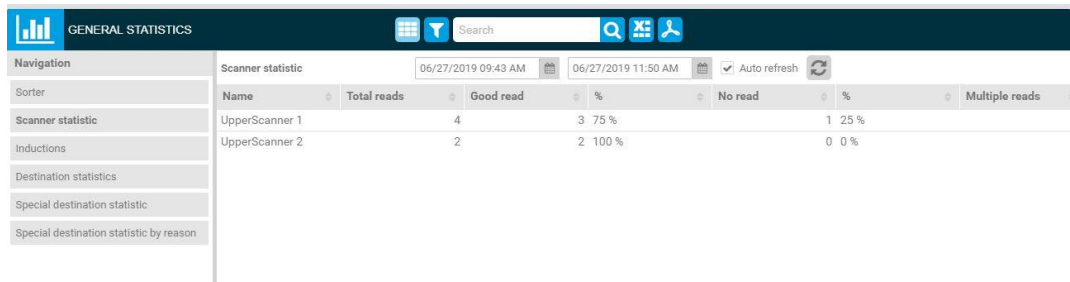
The scanner statistics gives an overview of the scanner result data.

Selection:

- Date/Time

The following statistic values are available:

Column	Description
Total Reads	Amount of all reads at the manual induction point.
Good Read %	Amount and % of good reads based on Total Reads
No Read	Amount of No Reads
No Read %	% of No Reads
Multiple Read %	% of Multiple Reads



The screenshot shows a web-based interface for 'GENERAL STATISTICS'. It features a navigation menu on the left with options like 'Scanner statistic', 'Inductions', 'Destination statistics', 'Special destination statistic', and 'Special destination statistic by reason'. The main area displays a table for 'Scanner statistic' with columns: Name, Total reads, Good read, %, No read, %, and Multiple reads. The data shows two scanners: 'UpperScanner 1' with 4 total reads (3 good, 1 no read) and 'UpperScanner 2' with 2 total reads (2 good, 0 no read). The interface also includes a search bar, a date range selector (06/27/2019 09:43 AM to 11:50 AM), and an 'Auto refresh' checkbox.

Name	Total reads	Good read	%	No read	%	Multiple reads
UpperScanner 1	4	3	75 %	1	25 %	
UpperScanner 2	2	2	100 %	0	0 %	

Figure 4-4 - Scanner statistical report

4.2.5 Destination Statistics

The chute statistics indicate the utilization of the chutes on a certain time range in a grid.

Selection:

- Date / Time

The following statistical values are available:

Column	Description
Count items	The total amount of discharged items will be displayed
Fault time	Duration of fault
Full time	Duration of full condition
Blocked time	Duration of the disabled/blocked condition

GENERAL STATISTICS								
Navigation	Destination statistics 06/27/2019 09:43 AM 06/27/2019 11:50 AM Auto refresh							
Sorter	Name	Count items	Full time	Jam time	Blocked time	Technical fault time	Operational fault time	
Scanner statistic	+1051.1	1	00:00:00	00:00:00	00:00:00	00:00:00	00:00:00	
Inductions	+1096.1	5	00:00:00	00:00:00	00:00:00	00:00:00	00:00:00	
Destination statistics								
Special destination statistic								
Special destination statistic by reason								

Figure 4-5 - Destination statistical report

4.2.6 Special Destination Statistics

This statistic displays the discharges grouped by discharge reason.

Selection:

- Date / Time

The following statistic values are available:

Column	Description
Total	The total amount of discharged items This is 100% basis for all the other percentages
OK	Items which have been successfully delivered (not to an exception chute)
Max. recirculation	Number of regular items that were sorted to a special destination due to max recirculation. These items are not counted into OK.
Max. recirc. special reason	Number of items that were already assigned to a special reason when they became max recirc. They will be counted for this counter as well as for the original special reason counter. This counter is not part of the Total value.
Following columns ...	These statistical counters are based on the configuration of exception chutes. A statistic counter is available for every possible exception discharge.

GENERAL STATISTICS									
Navigation	Special destination statistic 06/27/2019 09:43 AM 06/27/2019 11:50 AM 1h Auto refresh								
Sorter	Insert date	Total	OK	%	Max. recirculation	%	Max. recirc. special reason	%	Max.
Scanner statistic	06/27/2019 10:00:00 A 5	4	80 %	0	0 %	0	0 %	0	0
Inductions	06/27/2019 11:00:00 A 1	1	100 %	0	0 %	0	0 %	0	0
Destination statistics									
Special destination statistic									
Special destination statistic by reason									

Figure 4-6 - Special destinations report (discharge reason by time)

GENERAL STATISTICS									
Navigation	Special destination statistic by reason 06/27/2019 09:43 AM 06/27/2019 11:50 AM Auto refresh								
Sorter	Reason	Sum	%						
Scanner statistic	Total	6	100 %						
Inductions	OK	5	83.33 %						
Destination statistics	Max. recirculation	0	0 %						
Special destination statistic	Max. recirc. special rea:	0	0 %						
Special destination statistic by reason	Max. reci. carriers [%]	0	0 %						
	No read	1	16.67 %						
	Multi read	0	0 %						
	Stray item	0	0 %						
	Induction error	0	0 %						
	Insufficient Scanner Ga	0	0 %						
	Occupied twice	0	0 %						
	Discharge fault	0	0 %						
	Other faults	0	0 %						
	Disappeared	0	0 %						
	Destination not found	0	0 %						
	Dispatch	0	0 %						

Figure 4-7 - Special destination report (by reason)

Figure 4-8 - Sorter Controls Layout

4.3 Servers/VMs and Workstations

All PC hardware will be provided by DHL. BEUMER will provide minimum recommended hardware specifications in a separate document.

The sorter control (SC) system server will be installed in a virtual environment provided and maintained by DHL. BEUMER will provide minimum requirements in a separate document.

Two server instances are planned, one acting as master, the other as a cold backup. This implies that data (e.g. database) is automatically synchronized between the master and backup server instances, but the switch between master/backup is a manual process.

BGFusion workstations provide an on-site location to access BGFusion. These workstations may be placed where power is available and a network connection that can reach the SC server.

4.4 SC Application Handling

The sorter is equipped with a SC Master server and a SC Backup server. All processes are running as a service.

It is possible to start the SC on the SC Master or the SC Backup server. Other interfacing systems, like HOST, will always communicate with the server via the virtual IP address.

4.5 Redundancy

The SC server is a virtual machine that is provided by DHL. The two VM servers are in a cold backup set up. Data redundancy between the two SC servers is handled by the BEUMER cold backup procedures. The switching between master/backup servers remains a manual process to be handled by the customer when needed.

4.6 Database

It should be noted that the sorter control system uses the MS SQL database to create persistent disk-based data storage for transient data created and used while items are processed through the sortation system. Every event is registered, so the data amount is relatively small, but the number of reads/writes is proportional to the sort capacity of the system and can be quite high. Fast hard drives (15K RPM) or SSDs are deployed to achieve this. Additionally, we use a battery-backed-up read-write physical cache with traditional hard drives to secure the transaction capacity.

Sort transaction data is typically stored for a week, where after 'non-active' data is automatically purged. This is primarily done to facilitate the investigation of any reported anomalies in the sorting process.

4.7 Remote VPN

The principles of the remote control access are displayed in the following graphic.

Beumer/Crisplant VPN Remote Access Solution Topology

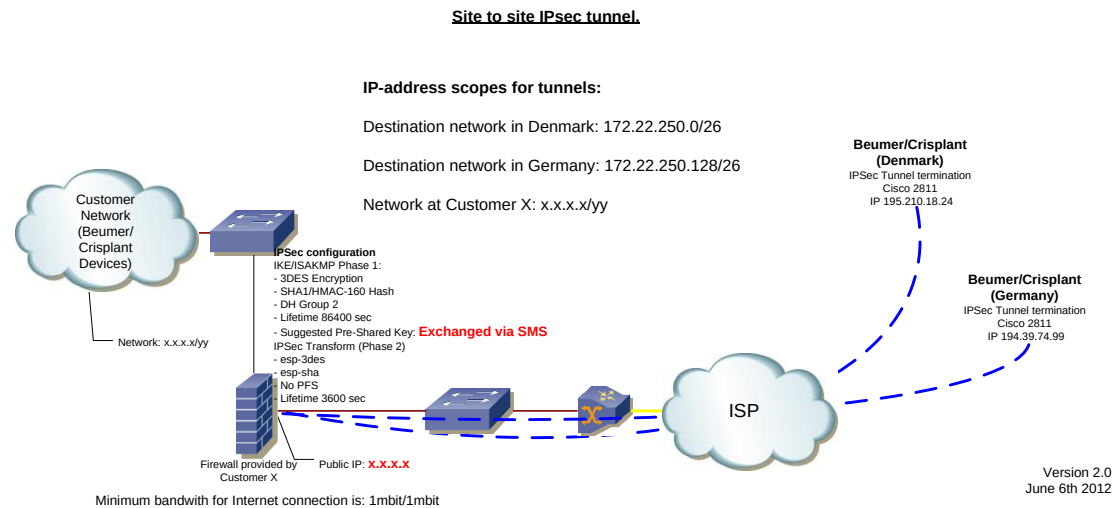


Figure 4-9 - Remote access scheme

Remote control access is a standard part of the system. Since a small problem may have a big impact on the system it is mandatory to be able to give quick and efficient remote support.

This is done by the standard remote support software 'NetSupportManager', NSM for the PC systems. All components are connected to a local Ethernet network. This network can be reached remotely from the BEUMER office via a VPN LAN to LAN connection.